DATA MINING PROJECT

Master in Data Science and Advanced Analytics

**NOVA Information Management School**

Universidade Nova de Lisboa

# Exploratory Data Analysis

## Amazing International Airlines Inc.

### Group 92

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# 1. ABSTRACT

This project presents an exploratory data analysis (EDA) of Amazing International Airlines Inc. (AIAI) loyalty-program data, intended to lay out the groundwork for an effective customer-segmentation strategy. Using three years of customer and flight-activity data, we performed a detailed examination across value-based metrics (Customer Lifetime Value and income), behavioral indicators (flight frequency, distance, activity patterns), and demographic attributes (education, location, status).

We uncovered key data-quality issues—such as missing income and CLV values, duplicated customer IDs, and inconsistencies among flight-related variables—which guided our data-cleaning and feature-engineering efforts.

Our insights include an uneven income distribution with peaks at very low and very high earners, a weak correlation between income and CLV, and a strong geographic concentration in three Canadian provinces. These findings form the analytical foundation for the upcoming segmentation phase, ensuring that ensuing customer clusters will be both meaningful and actionable for marketing and loyalty optimization.

# 2. INTRODUCTION

The modern airline industry is defined by fierce competition and a high stakes battle for customer loyalty. Carriers like **Amazing International Airlines Inc. (AIAI)** must move beyond generic strategies to design personalized services that drive retention and maximize lifetime value.

This project's main analytical goal is to provide a solid, data-driven foundation for consumer segmentation. To achieve this, we will undertake a comprehensive analysis of three years of loyalty membership and corresponding flight activity data. The segmentation strategy will not rely on a single dimension but will integrate insights from multiple perspectives: **Value-based** (grouping customers by their economic contribution or Customer Lifetime Value), **Behavioral** (analyzing purchasing habits, frequency, and travel patterns), and **Demographic** (categorizing customers by key attributes such as income and education). The generated segments are guaranteed to be unique, actionable, and comprehensively reflective of the AIAI client base due to our multi-perspective approach.

This segmentation is strategically vital for AIAI; in order to secure a lasting competitive advantage and maximize the potential revenue of the loyalty program, the resulting framework will offer practical insights to optimize marketing efficiency, customize the entire customer experience, and precisely adjust loyalty benefits.

# 3. DATA ANALYSIS

For this project we worked with two main sources: (1) the customer dataset, which contains one record per loyalty member with demographic, enrollment and value-related information, and (2) the flights dataset, which tracks monthly flight activity for each customer from January 2019 to December 2021 (up to 36 records per customer). The customer table is the master table, while the flights table provides a behavioral and temporal perspective that we will later use for segmentation.

### Data structure and granularity

The customer dataset contains unique customers and variables such as gender, education, location code, income, enrollment date, cancellation date, loyalty status and Customer Lifetime Value (CLV).

The flights dataset is at customer–month level and aggregates metrics like number of flights, flights with companions, total distance flown (DistanceKM) and points-related information. Each monthly row represents the total activity of that customer in that month, not individual trips. The column "Unnamed: 0" was excluded from the dataset as it represented an index-like field with no analytical or contextual relevance according to the available metadata.

**Data quality checks**

We first checked completeness and validity of the main attributes. Most fields in the customer dataset were populated, but two issues appeared:

- The column "Cancellation Date" has many missing values. We interpreted these as still-active customers and therefore created an additional binary variable "IsActive" instead of imputing dates.
- Around 20 customers had missing values in both "Income" and "CLV". Because both variables are important for a value-based analysis, we decided to keep these records at this stage, document them, and handle them later (drop or impute) before modeling.

Categorical variables (gender, education, marital status, location code, enrollment type, loyalty status) were checked for unexpected categories, typos, or mixed labels. No inconsistencies were found, so no recoding was necessary.

## Numerical consistency and duplicates

For numerical variables we applied basic business rules: no negative distances, no negative number of flights, and reasonable point values. During this step we detected a small number of duplicated loyalty IDs in the customer table. Because the same ID is also used in the flights table, keeping the duplicates would have doubled the linked flight history. Since the number of cases was small and the potential distortion high, we removed these duplicates. The income currency is not explicitly stated but is assumed to be in CAD based on dataset context.

**Logic validation**

After validating single variables, we checked for logical consistency across variables, for example:

- NumFlights = 0 but NumFlightsWithCompanions > 0
- NumFlights = 0 but DistanceKM > 0

These are logical errors, because a customer cannot have flight distance or companions without any flights. We flagged these records so that we can correct or exclude them in the data preparation phase. This is important to avoid bias in features that we will later use for clustering.

**Feature enrichment**

To make the data more informative for the later analysis, we created several derived variables:

- "IsActive" (based on the cancellation information)
- "CustomerTenureDays" (time between enrollment date and reference date)
- Customer-level flight metrics aggregated over the 36 months (total flights, average monthly flights, total distance, share of active months)
- A seasonal variable in the flight's dataset (Winter, Spring, Summer, Fall) based on the month of the flight, so that we can analyze seasonal travel behavior

All new variables were checked again for plausibility (e.g. no negative tenure, active customers without cancellation date).

**Exploratory profiling**
We then performed the EDA separately for customers and flights and later combined the two views. For the customer dataset we analyzed the distributions of income, CLV, loyalty status and enrollment type to understand potential drivers for a value- or behavior-based segmentation. For the flights dataset we analyzed activity over time and per season to see whether there are customers with stable vs. very seasonal flight patterns. We also looked specifically at customers with zero income, because this group looked unusual and might represent incomplete records or a special customer segment.

**Summary**
Overall, the data is suitable for segmentation, but we documented three issues that must be addressed in the next step:

1. Customers with missing values in both Income and CLV.
2. Records with logical inconsistencies between NumFlights and DistanceKM or companions.
3. Very low-activity customers that may distort distance- or points-based features.

Cleaning or excluding these cases in the data preparation phase will improve the stability and interpretability of the clustering in the next deliverable.

# 4. RESULTS

The exploratory data analysis (EDA) provided critical insights that directly informed the feature engineering and preprocessing stages of the clustering pipeline. Patterns observed across value-based, behavioral, and demographic dimensions helped determine which variables would be most relevant, how they should be transformed, and what data quality adjustments were necessary to ensure reliable segmentation.

From a **value-based perspective**, the limited correlation between income and Customer Lifetime Value (CLV) indicated that financial attributes alone cannot fully explain customer differences. This justified including behavioral and engagement-related variables in clustering, ensuring that segments capture both economic and experiential factors. Additionally, the extreme polarization in income and the presence of valid zero-income values influenced the decision to **normalize and scale income variables** rather than remove or impute them, preserving distributional diversity that may represent distinct market groups.

In the **behavioral dimension**, consistent seasonal patterns and the rise in flight activity between 2019 and 2021 supported the inclusion of features such as **total flights, average distance flown, and seasonal flight indicators can be observed**. These variables capture the contrast between steady, frequent flyers and seasonal travelers, which is crucial for differentiating engagement profiles. Companion flight patterns show little variation across loyalty statuses, so these metrics will be treated as secondary to tenure and flight consistency in the behavioral feature set.

For **demographic factors**, the heavy concentration of customers in Ontario, British Columbia, and Quebec revealed strong geographic bias, leading to the decision to **encode location as categorical variables** and later test its clustering relevance through variance analysis. Since education and income showed weak correlation, education level will be treated as a secondary attribute for cluster interpretation rather than a core segmentation driver.

Data validation outcomes also shaped preprocessing strategies. Duplicate loyalty IDs and inconsistencies—such as customers with positive flight distances but zero flights—were removed to eliminate distortions in activity-based features. Missing CLV and income values were retained for now but will be handled through **median or model-based imputation** during clustering preparation. Derived variables such as **IsActive** and **CustomerTenureDays** were retained as key indicators of engagement and program loyalty, directly linking customer lifecycle behavior to potential cluster differentiation.

These insights informed a targeted feature selection strategy that prioritized interpretable and high-impact variables. We retained the following attributes: Province/State, City, Location Code, Gender, Education, Marital Status, Income, Loyalty Status, Customer Lifetime Value (CLV), and the behavioral variables (total_flights, total_distance, total_flights_with_companions). Additionally, we engineered CustomerTenureDays and IsActive based on EnrollmentDateOpening and CancellationDate. Latitude and Longitude were preserved for potential geospatial analysis.

The following fields were excluded: 'Unnamed: 0' (index artifact), First/Last Name (personally identifiable information), Country (no variance, as all records correspond to Canada), Postal Code (too granular relative to City/Province), EnrollmentType (low informational value), and all points-related fields (high collinearity with flight and distance measures). Loyalty# was retained solely as a unique identifier and excluded from model training.

For the Flights dataset, we kept NumFlights, NumFlightsWithCompanions, DistanceKM, Year, Month, and Season, as these variables capture behavioral and seasonal patterns. Remaining fields were removed due to redundancy or limited relevance for clustering.

At this stage, no detailed standardization or encoding procedures were applied, since Deliverable 1 focused on exploratory analysis and feature selection. A comprehensive preprocessing pipeline—covering standardization of continuous variables (e.g., Income, CLV, distance metrics, CustomerTenureDays) and one-hot encoding of categorical features (e.g., Province/State, Gender, Education, Loyalty Status, Season)—will be implemented in Deliverable 2 to ensure all features are appropriately scaled and formatted for clustering.

In summary, the exploratory findings shaped a clear, data-driven rationale for preprocessing and feature engineering. The relationships uncovered between CLV, income, geography, and flight behavior ensure that the next phase—clustering and segmentation—will be grounded in meaningful variables that reflect real customer diversity rather than statistical noise. This alignment between business insight and analytical preparation strengthens the foundation for building actionable, high-precision customer segments that can directly inform loyalty, marketing, and retention strategies.

# 5. CONCLUSION

The exploratory phase of this project established a robust analytical groundwork for Phase 2 — clustering and segmentation modelling. Through systematic data-quality validation, feature enrichment, and profiling, we ensure that the dataset accurately represents Amazing International Airlines Inc. (AIAI)-s customer base while addressing inconsistencies that could bias results. Our findings highlight the diversity of the customer population across three perspectives:

**Value-based:** The strong polarization in income levels and the weak relationship between income and CLV suggest that purely financial variables may not fully explain customer behavior or loyalty potential.

**Behavioral:** Seasonal and frequency-based variations in flight activity indicate that engagement and travel behavior are critical segmentation drivers, particularly for differentiating between consistent and opportunistic flyers.

**Demographic:** The dominance of Ontario, British Columbia, and Quebec in both customer volume and CLV underscores the importance of location-specific marketing strategies.

Based on these insights, our clustering approach will integrate behavioral, value-based, and demographic features to capture both transactional and attitudinal differences. We plan to preprocess the data through normalization, imputation of missing values (where appropriate), and removal of logically inconsistent records. Feature selection will emphasize interpretable variables such as total flights, average distance flown, CLV, and tenure days.

The expected segmentation outcome is the identification of distinct customer groups with unique value and engagement profiles—ranging from high-value frequent travellers to low-engagement occasional users—allowing AIAI to tailor marketing campaigns, reward structures, and service personalization effectively. Ultimately, this EDA phase confirms that the available data is suitable for clustering and provides a clear roadmap for building actionable customer segments in the next stage.

Additionally, we have documented data-cleaning protocols and feature-engineering decisions so that the segmentation process remains transparent and reproducible. We also reviewed potential bias sources—such as geographic over-representation—and proposed stratified sampling to ensure that the clustering results reflect the full breadth of the customer base. With this analytical foundation in place, the next phase will focus on algorithm selection (e.g., k-means, hierarchical clustering), validation of cluster stability, and generation of actionable personas, and campaign-ready profiles for each segment.

# 6. ANNEX

## 6.1 ANNEX AI USAGE STATEMENT

AI tools were used to support the project in limited and well-defined ways.
ChatGPT was used for coding syntax assistance (mainly for pandas operations, data cleaning logic, and Python debugging), as well as for help with writing structure, text refinement, and proofreading.
GitHub Copilot was used within Visual Studio Code to suggest short code completions and function syntax during implementation.
All analytical insights, business interpretations, and strategic recommendations are the result of the group's independent analysis and reasoning.

## 6.2 ANNEX CONTRIBUTION STATEMENT

**Duarte Manuel André Gomes (Student ID: 20250017)**

- Business Understanding
- Exploratory Data Analysis
- Data Visualization
- Demographic Segmentation
- Results Section

**Esra Salhi (Student ID: 20250537)**

- Business Understanding
- Behavioral Segmentation
- Introduction section
- Conclusion section
- Poster Design

**Mehmet Karaca (Student ID: 20250344)**

- Data Understanding
- Missing Values and Data Validity Checks
- Feature Engineering
- Exploratory Data Analysis
- Value Based Segmentation
- Data Analysis Section
- Poster Design

## 6.3 ANNEX RESPONSIBILITY STATEMENT

We, the group members listed above, certify that this report represents our original analytical work and interpretations. While AI tools were used as specified above, all insights, conclusions, and recommendations are the result of our independent analysis and critical thinking. We take full responsibility for the accuracy and quality of this submission.

# 7. BIBLIOGRAPHICAL REFERENCES

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